Arduino for Embedded Systems - Beginner Level

Lesson 6: Using Libraries: LCD and Temperature Sensor

Using Libraries: LCD and Temperature Sensor – Study Notes

Key Concepts

1. **Including and Initialising Libraries** – `LiquidCrystal` for the LCD and `DHT` for the DHT11 temperature/humidity sensor.

2. **Pin Connections & Wiring** – Mapping Arduino pins to the LCD (RS, EN, D4 D7) and to the DHT11 data pin.

3. **Reading Sensor Data** – Using `dht.readTemperature()` and `dht.readHumidity()` and handling read errors.

4. **Displaying Text on the LCD** – Positioning the cursor, clearing the screen, and formatting numeric values.

5. **Timing & Refresh Rate** – Updating the display at a sensible interval (e.g., every 2 /s) without blocking other code.

Summary

In this lesson you learn how to combine two of the most common Arduino peripherals: a 16×2 character LCD (driven by the **LiquidCrystal** library) and a DHT11 temperature/humidity sensor (controlled by the **DHT** library). After installing the libraries, you include them in your sketch, create objects that represent the hardware, and initialise them in `setup()`.

The main loop repeatedly reads the temperature (in °C) and humidity (%) from the DHT11, checks for read errors, and prints the values to the LCD. By using `lcd.setCursor()` you can place the temperature on the first line and the humidity on the second line, giving a clear, real time readout. A short delay (or a non blocking timing scheme) prevents the sensor from being polled too quickly and keeps the display legible.

Important Points

- **Library Installation**

- Open the Arduino IDE !’ *Sketch !’ Include Library !’ Manage Libraries…*

- Search for **LiquidCrystal** (bundled) and **DHT sensor library** (by Adafruit) and install.

- **Wiring the LCD (4 bit mode)**

LCD Pin	Arduino Pin
RS	12
EN	11

D4	5
D5	4
D6	3
D7	2
VSS/GND	GND
VDD/VCC	5 /V
VO (contrast)	middle of a 10 /k potentiometer (ends to 5 /V & GND)
LED+ (backlight)	5 /V (via 220 /:’’ À
LED-	GND

- **Wiring the DHT11**

- VCC !’ 5 /V, GND !’ GND, Data !’ any digital pin (e.g., **7**) with a 10 /k /:’ VÆÂ W &W6—7F÷" Fò R õ`.

- **Object Declaration**

```cpp
#include <LiquidCrystal.h>
#include <DHT.h>
const int rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
#define DHTPIN 7
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
```

- **Reading the Sensor**

```cpp
float temperature = dht.readTemperature();   // Celsius
float humidity    = dht.readHumidity();
if (isnan(temperature) || isnan(humidity)) {
  // handle error – often just skip this cycle
}
```

- **Displaying Values**

```cpp
lcd.clear();
lcd.setCursor(0,0);
lcd.print("Temp: ");
lcd.print(temperature, 1);   // one decimal place
lcd.print((char)223);        // degree symbol
lcd.print("C");
lcd.setCursor(0,1);
lcd.print("Humid: ");
lcd.print(humidity, 1);
lcd.print("%");
```

• **Timing** – Use `millis()` for non blocking updates:
```cpp
const unsigned long interval = 2000; // 2 seconds
unsigned long previous = 0;
void loop() {
    if (millis() - previous >= interval) {
        previous = millis();
        // read sensor & update LCD
    }
}
```
• **Common Pitfalls**
- Forgetting the pull up resistor on the DHT data line.
- Using the wrong LCD pin order – results in garbled characters.
- Updating the LCD too fast (the DHT11 needs ~1 /s between reads).

Examples

Example 1 – Basic Temperature & Humidity Display

```cpp
/*  LCD + DHT11 demo – prints temperature and humidity every 2 seconds */
#include <LiquidCrystal.h>
#include <DHT.h>
const int rs = 12, en = 11, d4 = 5, d5 = 4, d6 = 3, d7 = 2;
LiquidCrystal lcd(rs, en, d4, d5, d6, d7);
#define DHTPIN 7
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
const unsigned long UPDATE_INTERVAL = 2000;
unsigned long lastUpdate = 0;
void setup() {
  lcd.begin(16, 2);          // 16 columns, 2 rows
  lcd.print("Initializing...");
  dht.begin();
  delay(1000);
  lcd.clear();
}
void loop() {
  if (millis() - lastUpdate >= UPDATE_INTERVAL) {
    lastUpdate = millis();
    float t = dht.readTemperature();
    float h = dht.readHumidity();
```

```
    if (isnan(t) || isnan(h)) {
      lcd.clear();
      lcd.print("Read error");
      return;
    }
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Temp: ");
    lcd.print(t, 1);
    lcd.print((char)223);
    lcd.print("C");
    lcd.setCursor(0, 1);
    lcd.print("Humid: ");
    lcd.print(h, 1);
    lcd.print("%");
  }
}
```

Explanation: The sketch sets up the LCD and DHT objects, then every two seconds reads the sensor and writes formatted values to the two rows of the LCD. Errors are shown briefly.

Example 2 – Non Blocking Multi Task Sketch

```cpp
/*  Demonstrates using millis() for sensor read & LCD update
    while also blinking an LED on pin 13. */
#include <LiquidCrystal.h>
#include <DHT.h>
LiquidCrystal lcd(12, 11, 5, 4, 3, 2);
#define DHTPIN 7
#define DHTTYPE DHT11
DHT dht(DHTPIN, DHTTYPE);
const unsigned long SENSOR_INTERVAL = 2000;
const unsigned long LED_INTERVAL    = 500;
unsigned long lastSensor = 0, lastLED = 0;
bool ledState = false;
void setup() {
  pinMode(13, OUTPUT);
  lcd.begin(16, 2);
  dht.begin();
  lcd.print("Ready");
}
void loop() {
```

```
  // ---- LED blink (independent) ----
  if (millis() - lastLED >= LED_INTERVAL) {
    lastLED = millis();
    ledState = !ledState;
    digitalWrite(13, ledState);
  }
  // ---- Sensor read & LCD update ----
  if (millis() - lastSensor >= SENSOR_INTERVAL) {
    lastSensor = millis();
    float t = dht.readTemperature();
    float h = dht.readHumidity();
    lcd.clear();
    if (isnan(t) || isnan(h)) {
      lcd.print("Sensor error");
    } else {
      lcd.setCursor(0,0);
      lcd.print("T:");
      lcd.print(t,1);
      lcd.print((char)223);
      lcd.print("C ");
      lcd.setCursor(0,1);
      lcd.print("H:");
      lcd.print(h,1);
      lcd.print("%");
    }
  }
}
```

Explanation: Shows how to keep the Arduino responsive by avoiding `delay()`. The LED blinks every 500 /ms while the sensor/LCD updates every 2 /s.

Quick Review

1. **What two libraries are required to drive a 16×2 LCD and a DHT11 sensor?**
2. **Why is a pull up resistor needed on the DHT11 data line?**
3. **Write the line of code that creates a `LiquidCrystal` object for pins RS=12, EN=11, D4 D7=5,4,3,2.**
4. **How can you display the degree symbol (°) on the LCD?**
5. **Explain why using `millis()` is preferred over `delay()` when you need to read a sensor and update an LCD.**

Further Reading

- **Advanced LCD Features** – custom characters, scrolling text, and using I²C LCD backpacks

(`LiquidCrystal_I2C`).

• **Other DHT Sensors** – DHT22/AM2302 differences, higher accuracy, and library usage.

• **Non Blocking Programming** – state machines, `Ticker` library, and cooperative multitasking on Arduino.

• **Power Management** – putting the Arduino to sleep between sensor reads to save battery.

• **Serial Debugging** – printing sensor values to the Serial Monitor for troubleshooting.